

For example, controls experts will perform a six-degree-of-freedom analysis to ensure that the designer's estimate for the size of the control surfaces is adequate for control under all conditions required by design specifications. If not, they will instruct the designer as to how much each control surface must be expanded. If a larger aileron is required, the designer must ensure that it can be incorporated into the design without adversely affecting something else, such as the flaps or the landing gear.

The end product of all this will be an aircraft design that can be confidently passed to the preliminary design phase, as previously discussed. While further changes should be expected during preliminary design, major revisions will not occur if the conceptual design effort has been successful.

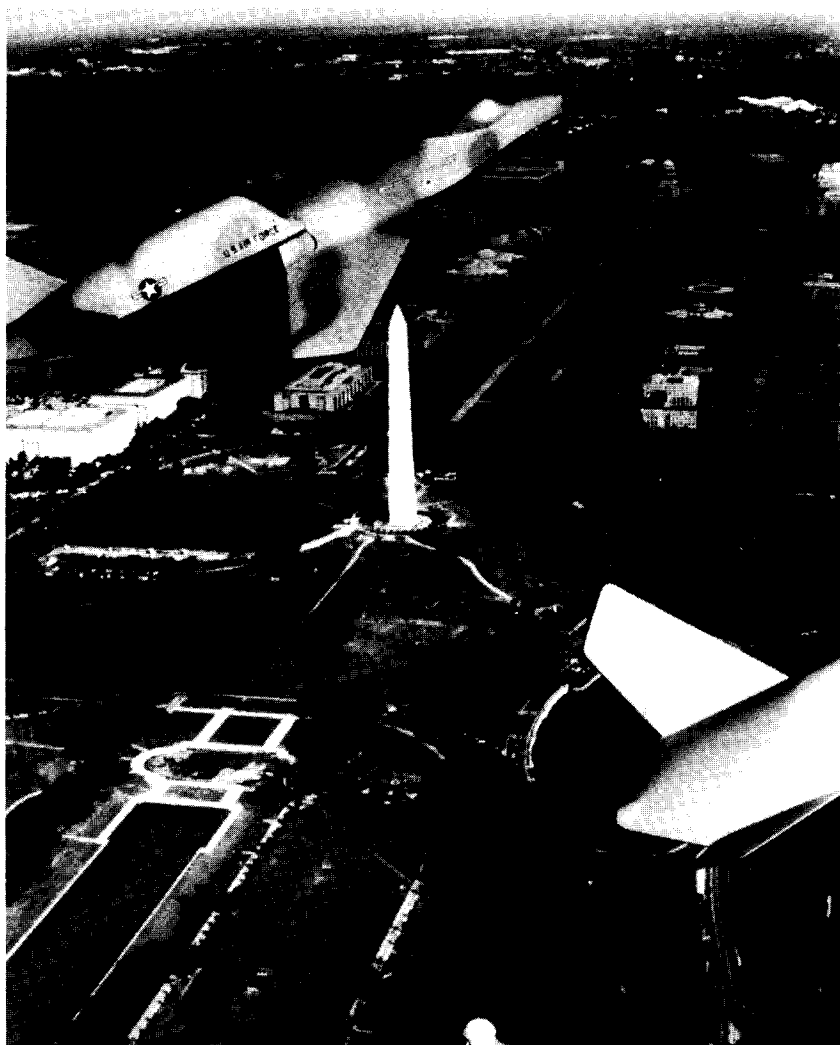


Photo by Bruce Frisch

Reverse Installation Vectored Engine Thrust ("RIVET") Supersonic VSTOL
Concept Model

3 SIZING FROM A CONCEPTUAL SKETCH

3.1 INTRODUCTION

There are many levels of design procedure. The simplest level just adopts past history. For example, if you need an immediate estimate of the takeoff weight of an airplane to replace the Air Force F-15 fighter, use 44,500 lb. That is what the F-15 weighs, and is probably a good number to start with.

To get the "right" answer takes several years, many people, and lots of money. Actual design requirements must be evaluated against a number of candidate designs, each of which must be designed, analyzed, sized, optimized, and redesigned any number of times.

Analysis techniques include all manner of computer code as well as correlations to wind-tunnel and other tests. Even with this extreme level of design sophistication, the actual airplane when flown will never exactly match predictions.

In between these extremes of design and analysis procedures lie the methods used for most conceptual design activities. As an introduction to the design process, this chapter presents a quick method of estimating takeoff weight from a conceptual sketch.

The simplified sizing method presented in this chapter can only be used for missions which do not include any combat or payload drops. While admittedly crude, this method introduces all of the essential features of the most sophisticated design by the major aerospace manufacturers. In a later chapter, the concepts introduced here will be expanded to a sizing method capable of handling all types of mission.

3.2 TAKEOFF-WEIGHT BUILDUP

"Design takeoff gross weight" is the total weight of the aircraft as it begins the mission for which it was designed. This is not necessarily the same as the "maximum takeoff weight." Many military aircraft can be overloaded beyond design weight but will suffer a reduced maneuverability. Unless specifically mentioned, takeoff gross weight, or " W_0 ," is assumed to be the design weight.

Design takeoff gross weight can be broken into crew weight, payload (or passenger) weight, fuel weight, and the remaining (or "empty") weight. The empty weight includes the structure, engines, landing gear, fixed equipment, avionics, and anything else not considered a part of crew, payload, or fuel. Equation (3.1) summarizes the takeoff-weight buildup.

$$W_0 = W_{\text{crew}} + W_{\text{payload}} + W_{\text{fuel}} + W_{\text{empty}} \quad (3.1)$$

The crew and payload weights are both known since they are given in the design requirements. The only unknowns are the fuel weight and empty weight. However, they are both dependent on the total aircraft weight. Thus an iterative process must be used for aircraft sizing.

To simplify the calculation, both fuel and empty weights can be expressed as fractions of the total takeoff weight, i.e., (W_f/W_0) and (W_e/W_0) . Thus Eq. (3.1) becomes:

$$W_0 = W_{\text{crew}} + W_{\text{payload}} + \left(\frac{W_f}{W_0}\right)W_0 + \left(\frac{W_e}{W_0}\right)W_0 \quad (3.2)$$

This can be solved for W_0 as follows:

$$W_0 - \left(\frac{W_f}{W_0}\right)W_0 - \left(\frac{W_e}{W_0}\right)W_0 = W_{\text{crew}} + W_{\text{payload}} \quad (3.3)$$

$$W_0 = \frac{W_{\text{crew}} + W_{\text{payload}}}{1 - (W_f/W_0) - (W_e/W_0)} \quad (3.4)$$

Now W_0 can be determined if (W_f/W_0) and (W_e/W_0) can be estimated. These are described below.

3.3 EMPTY-WEIGHT ESTIMATION

The empty-weight fraction (W_e/W_0) can be estimated statistically from historical trends as shown in Fig. 3.1, developed by the author from data taken from Ref. 1 and other sources. Empty-weight fractions vary from about 0.3 to 0.7, and diminish with increasing total aircraft weight.

As can be seen, the type of aircraft also has a strong effect, with flying boats having the highest empty-weight fractions and long-range military aircraft having the lowest. Flying boats are heavy because they need to carry extra weight for what amounts to a boat hull. Notice also that different types of aircraft exhibit different slopes to the trend lines of empty-weight fraction vs takeoff weight.

Table 3.1 presents statistical curve-fit equations for the trends shown in Fig. 3.1. Note that these are all exponential equations based upon takeoff gross weight. The exponents are small negative numbers, which indicates that the empty weight fractions decrease with increasing takeoff weight, as shown by the trend lines in Fig. 3.1. The differences in exponents for different types of aircraft reflect the different slopes to the trend lines, and imply that some types of aircraft are more sensitive in sizing than others.

A variable-sweep wing is heavier than a fixed wing, and is accounted for at this initial stage of design by multiplying the empty-weight fraction as determined from the equations in Table 3.1 by about 1.04.

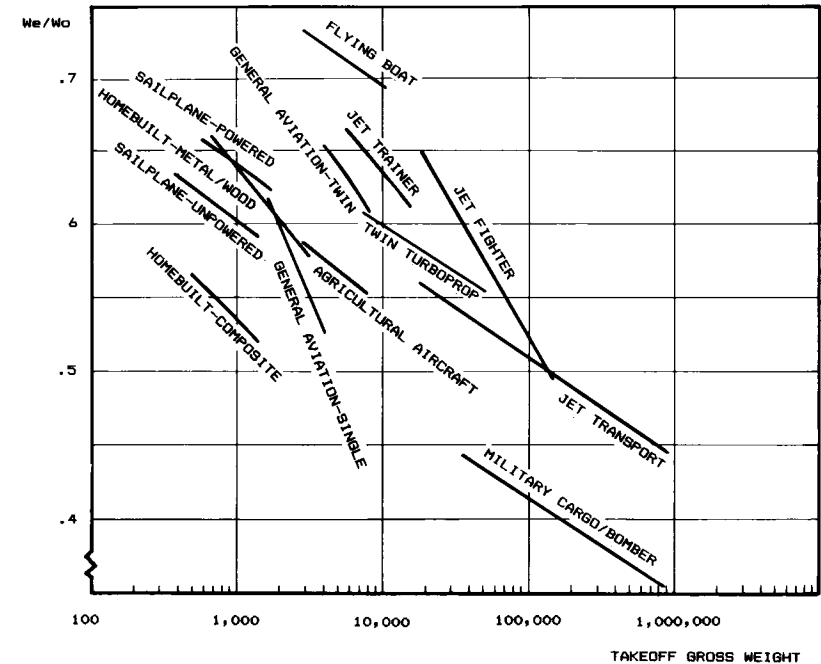


Fig. 3.1 Empty weight fraction trends.

Table 3.1 Empty weight fraction vs W_0

$W_e/W_0 = A W_0^C K_{vs}$	A	C
Sailplane—unpowered	0.86	-0.05
Sailplane—powered	0.91	-0.05
Homebuilt—metal/wood	1.19	-0.09
Homebuilt—composite	0.99	-0.09
General aviation—single engine	2.36	-0.18
General aviation—twin engine	1.51	-0.10
Agricultural aircraft	0.74	-0.03
Twin turboprop	0.96	-0.05
Flying boat	1.09	-0.05
Jet trainer	1.59	-0.10
Jet fighter	2.34	-0.13
Military cargo/bomber	0.93	-0.07
Jet transport	1.02	-0.06

K_{vs} = variable sweep constant = 1.04 if variable sweep
= 1.00 if fixed sweep

Advanced composite materials such as graphite-epoxy are replacing aluminum in a number of new designs. There have not yet been enough composite aircraft flown to develop statistical equations. Based on a number of design studies, the empty-weight fraction for other types of composite aircraft can be estimated at this stage by multiplying the statistical empty-weight fraction by 0.95.

“Composite” homebuilt aircraft are typically of fiberglass-epoxy construction rather than an advanced composite material. The statistically estimated empty weight fraction for fiberglass-epoxy composite homebuilts is approximately 0.85 times the metal homebuilt empty-weight fraction ($0.99/1.19 = 0.85$). However, this is not due to the material used for construction as much as the different design philosophies concerning ease of manufacture, passenger comfort, maintenance accessibility, and similar factors.

3.4 FUEL-FRACTION ESTIMATION

Only part of the aircraft's fuel supply is available for performing the mission (“mission fuel”). The other fuel includes reserve fuel as required by civil or military design specifications, and also includes “trapped fuel,” which is the fuel which cannot be pumped out of the tanks.

The required amount of mission fuel depends upon the mission to be flown, the aerodynamics of the aircraft, and the engine's fuel consumption. The aircraft weight during the mission affects the drag, so the fuel used is a function of the aircraft weight.

As a first approximation, the fuel used can be considered to be proportional to the aircraft weight, so the fuel fraction (W_f/W_0) is approximately independent of aircraft weight. Fuel fraction can be estimated based on the mission to be flown using approximations of the fuel consumption and aerodynamics.

Mission Profiles

Typical mission profiles for various types of aircraft are shown in Fig. 3.2. The Simple Cruise mission is used for many transport and general-aviation designs, including homebuilts. The aircraft is sized to provide some required cruise range.

For safety you would be wise to carry extra fuel in case your intended airport is closed, so a loiter of typically 20–30 min is added. Alternatively, additional range could be included, representing the distance to the nearest other airport or some fixed number of minutes of flight at cruise speed (the FAA requires 30 min of additional cruise fuel for general-aviation aircraft).

Other missions are more complex. The typical Air Superiority mission includes a cruise out, a combat consisting of either a certain number of turns or a certain number of minutes at maximum power, a weapons drop, a cruise back, and a loiter. The weapons drop refers to the firing of gun and missiles, and is often left out of the sizing analysis to insure that the aircraft has enough fuel to return safely if the weapons aren't used. Note that the second cruise segment is identical to the first, indicating that the aircraft must return to its base at the end of the mission.

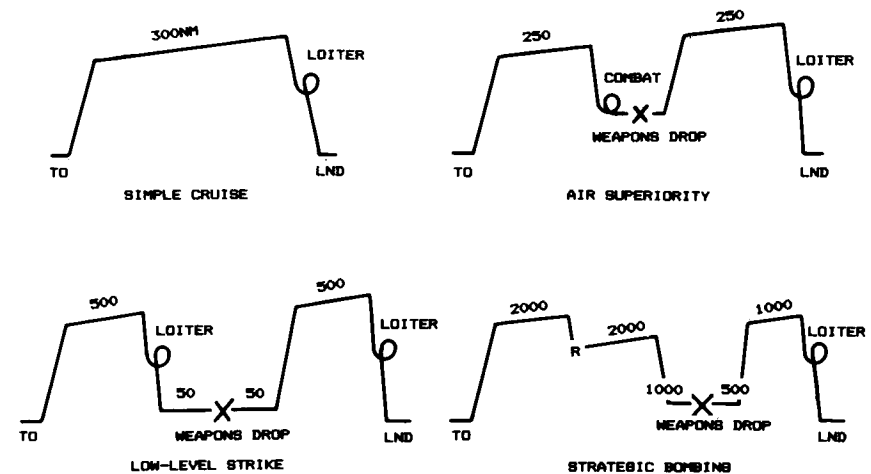


Fig. 3.2 Typical mission profiles for sizing.

The Low-Level Strike mission includes “dash” segments that must be flown at just a few hundred feet off the ground. This is to improve the survivability of the aircraft as it approaches its target. Unfortunately, the aerodynamic efficiency of an aircraft, expressed as “lift-to-drag ratio” (L/D), is greatly reduced during low-level, high-speed flight, as is the engine efficiency. The aircraft may burn almost as much fuel during the low-level dash segment as it burns in the much-longer cruise segment.

The Strategic Bombing mission introduces another twist. After the initial cruise, a refueling segment occurs, as indicated by an “R.” Here the aircraft meets up with a tanker aircraft such as an Air Force KC-135 and receives some quantity of fuel. This enables the bomber to achieve far more range, but adds to the overall operating cost because a fleet of tanker aircraft must be dedicated to supporting the bombers.

Also note that the bomber in this typical strategic mission will fly at low level as it nears the target area to improve its chances of survival. Earlier bombers such as the B-52 were originally designed to cruise at high altitudes throughout the mission.

Another difference in this strategic mission is the fact that the return cruise range is far shorter than the outbound range. This is necessary because of the extreme range required. If the aircraft were sized to return to its original base, it would probably weigh several million pounds. Instead, it is assumed that strategic bombers will land on bases in friendly countries for refueling after completion of their mission.

These are merely typical missions, and the ranges shown are just examples. When an aircraft is designed, the actual mission profile and ranges will be provided by the customer or determined by operational analysis methods beyond the scope of this book.

In addition to the mission profile, requirements will be established for a number of performance parameters such as takeoff distance, maneuverabil-

ity, and climb rates. These are ignored in the simplified sizing method of this chapter, but will be discussed in detail later.

Mission Segment Weight Fractions

For analysis, the various mission segments, or “legs,” are numbered, with zero denoting the start of the mission. Mission leg “one” is usually engine warmup and takeoff for first-order sizing estimation. The remaining legs are sequentially numbered.

For example, in the simple cruise mission the legs could be numbered as (1) warmup and takeoff, (2) climb, (3) cruise, (4) loiter, and (5) land (see the example mission at the end of this chapter).

In a similar fashion, the aircraft weight at each part of the mission can be numbered. Thus, W_0 is the beginning weight (“takeoff gross weight”).

For the simple cruise mission, W_1 would be the weight at the end of the first mission segment, which is the warmup and takeoff. W_2 would be the aircraft weight at the end of the climb. W_3 would be the weight after cruise, and W_4 after loiter. Finally, W_5 would be the weight at the end of the landing segment, which is also the end of the total mission.

During each mission segment, the aircraft loses weight by burning fuel (remember that our simple sizing method doesn’t permit missions involving a payload drop). The aircraft weight at the end of a mission segment divided by its weight at the beginning of that segment is called the “mission segment weight fraction.” This will be the basis for estimating the required fuel fraction for initial sizing.

For any mission segment “ i ,” the mission segment weight fraction can be expressed as (W_i/W_{i-1}) . If these weight fractions can be estimated for all of the mission legs, they can be multiplied together to find the ratio of the aircraft weight at the end of the total mission, W_x (assuming “ x ” segments altogether) divided by the initial weight, W_0 . This ratio, W_x/W_0 , can then be used to calculate the total fuel fraction required.

These mission segment weight fractions can be estimated by a variety of methods. For our simplified form of initial sizing, the types of mission leg will be limited to warmup and takeoff, climb, cruise, loiter, and land. As previously mentioned, mission legs involving combat, payload drop, and refuel are not permitted in this simplified sizing method but will be discussed in a later chapter.

The warmup, takeoff, and landing weight-fractions can be estimated historically. Table 3.2 gives typical historical values for initial sizing. These

Table 3.2 Historical mission segment weight fractions

	(W_i/W_{i-1})
Warmup and takeoff	0.970
Climb	0.985
Landing	0.995

values can vary somewhat depending on aircraft type, but the averaged values given in the table are reasonable for initial sizing.

In our simple sizing method we ignore descent, assuming that the cruise ends with a descent and that the distance traveled during descent is part of the cruise range.

Cruise-segment mission weight fractions can be found using the Breguet range equation (derived in Chapter 17):

$$R = \frac{V}{C} \frac{L}{D} \ln \frac{W_{i-1}}{W_i} \quad (3.5)$$

or

$$\frac{W_i}{W_{i-1}} = \exp \frac{-RC}{V(L/D)} \quad (3.6)$$

where

- R = range
- C = specific fuel consumption
- V = velocity
- L/D = lift-to-drag ratio

Loiter weight fractions are found from the endurance equation (also derived in Chapter 17):

$$E = \frac{L/D}{C} \ln \frac{W_{i-1}}{W_i} \quad (3.7)$$

or

$$\frac{W_i}{W_{i-1}} = \exp \frac{-EC}{L/D} \quad (3.8)$$

where E = endurance or loiter time.

(Note: It is very important to use consistent units! Also note that C and L/D vary with speed and altitude. Furthermore, C varies with throttle setting, and L/D varies with aircraft weight. These will be discussed in detail in later chapters.)

Specific Fuel Consumption

Specific fuel consumption (“SFC” or simply “ C ”) is the rate of fuel consumption divided by the resulting thrust. For jet engines, specific fuel consumption is usually measured in pounds of fuel per hour per pound of thrust [(lb/hr)/lb, or 1/hr]. Figure 3.3 shows SFC vs Mach number.

Propeller engine SFC is normally given as C_{bhp} , the pounds of fuel per hour to produce one horsepower at the propeller shaft (or one “brake horsepower”: bhp = 550 ft-lb/s).

A propeller thrust SFC equivalent to the jet-engine SFC can be calculated. The engine produces thrust via the propeller, which has an efficiency η_p

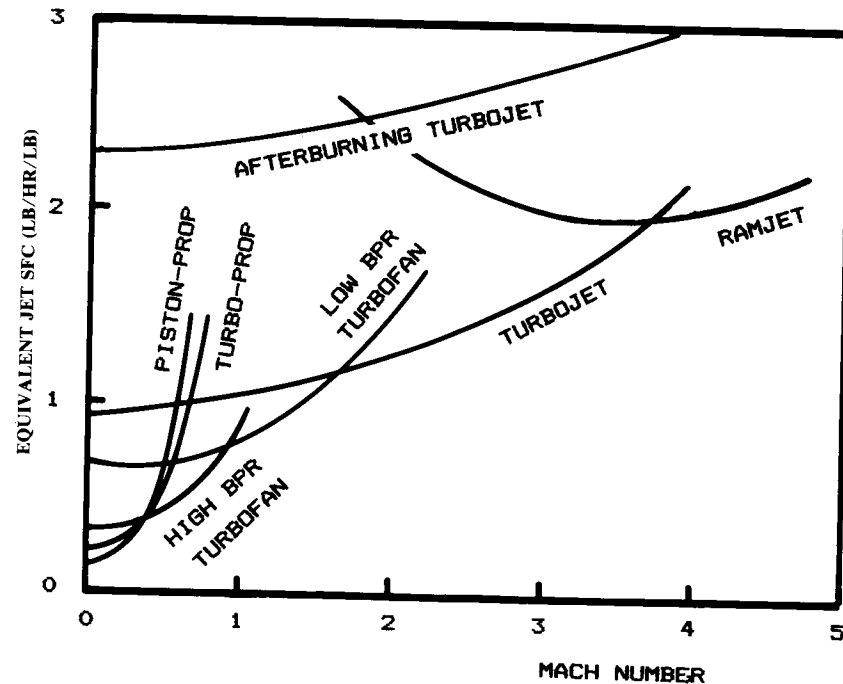


Fig. 3.3 Specific fuel consumption trends.

defined as thrust power output per horsepower input [Eq. (3.9)]. The 550 term assumes that V is in feet per second.

$$\eta_p = \frac{TV}{550 \text{ hp}} \quad (3.9)$$

Equation (3.10) shows the derivation of the equivalent-thrust SFC for a propeller-driven aircraft. Note that for a propeller aircraft, the thrust and the SFC are a function of the flight velocity. For a typical aircraft with a propeller efficiency of about 0.8, one horsepower equals one pound of thrust at about 440 ft/s, or about 260 knots.

$$C = \frac{W_f/\text{time}}{\text{thrust}} = C_{bhp} \frac{V}{550 \eta_p} \quad (3.10)$$

Table 3.3 provides typical SFC values for jet engines, while Table 3.4 provides typical C_{bhp} and η_p values for propeller engines. These can be used for rough initial sizing. In later chapters more detailed procedures for calculating these values, which change as a function of altitude, velocity, and power setting, will be presented.

Table 3.3 Specific fuel consumption (C)

Typical jet SFC's	Cruise	Loiter
Pure turbojet	0.9	0.8
Low-bypass turbofan	0.8	0.7
High-bypass turbofan	0.5	0.4

Table 3.4 Propeller specific fuel consumption (C_{bhp})

Propeller: $C = C_{bhp} V / (550 \eta_p)$	Cruise	Loiter
Typical C_{bhp} and η_p		
Piston-prop (fixed pitch)	0.4/0.8	0.5/0.7
Piston-prop (variable pitch)	0.4/0.8	0.5/0.8
Turboprop	0.5/0.8	0.6/0.8

L/D Estimation

The remaining unknown in both range and loiter equations is the L/D , or lift-to-drag ratio, which is a measure of the design's overall aerodynamic efficiency. Unlike the parameters estimated above, the L/D is highly dependent upon the configuration arrangement. At subsonic speeds L/D is most directly affected by two aspects of the design: wing span and wetted area.

In level flight, the lift is known. It must equal the aircraft weight. Thus, L/D is solely dependent upon drag.

The drag at subsonic speeds is composed of two parts. "Induced" drag is the drag caused by the generation of lift. This is primarily a function of the wing span.

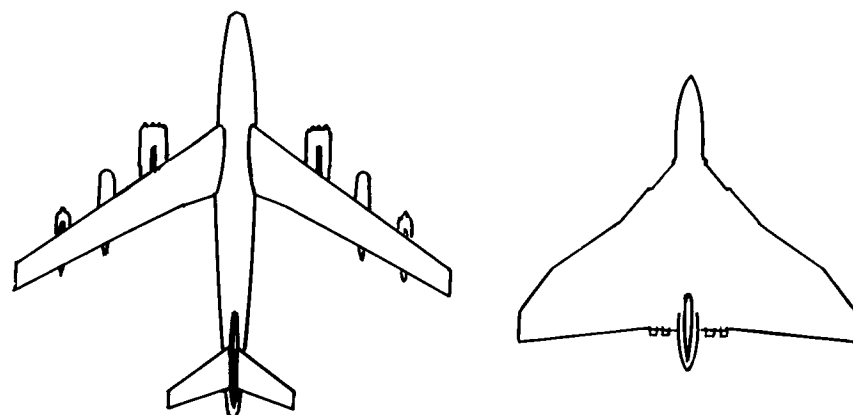
"Zero-lift," or "parasite" drag is the drag which is not related to lift. This is primarily skin-friction drag, and as such is directly proportional to the total surface area of the aircraft exposed ("wetted") to the air.

The "aspect ratio" of the wing has historically been used as the primary indicator of wing efficiency. Aspect ratio is defined as the square of the wing span divided by the wing reference area. For a rectangular wing the aspect ratio is simply the span divided by chord.

Aspect ratios range from under one for re-entry lifting bodies to over thirty for sailplanes. Typical values range between three and eight. For initial design purposes, aspect ratio can be selected from historical data. For final determination of the best aspect ratio, a trade study as discussed in Chapter 19 should be conducted.

Aspect ratio could be used to estimate subsonic L/D , but for one major problem. The parasite drag is not a function of just the wing area, as expressed by aspect ratio, but also of the aircraft's total wetted area.

Figure 3.4 shows two widely different aircraft concepts, both designed to perform the same mission of strategic bombing. The Boeing B-47 features a conventional approach. With its aspect ratio of over 9, it is not surprising



	B-47	AVRO VULCAN
S reference	1430	3446
S wetted	11300	9600
SPAN	116	90
Swet/Sref	7.9	2.8
ASPECT RATIO	9.4	3.0
WETTED ASPECT RATIO	1.2	1.1
L/D max	17.2	17.0

Fig. 3.4 Does aspect ratio predict drag?

that it attains an L/D of over 17. On the other hand, the AVRO Vulcan bomber has an aspect ratio of only 3, yet it attains almost exactly the same L/D .

The explanation for this curious outcome lies in the actual drivers of L/D as discussed above. Both aircraft have about the same wing span, and both have about the same wetted areas, so both have about the same L/D . The aspect ratio of the B-47 is higher not because of a greater wing span, but because of a smaller wing area. However, this reduced wing area is offset by the wetted area of the fuselage and tails.

This is illustrated by the ratios of wetted area to wing reference area (S_{wet}/S_{ref}). While the AVRO design has a total wetted area of less than three times the wing area, the Boeing design has a wetted area of eight times the wing area.

This wetted-area ratio can be used, along with aspect ratio, as a more reliable early estimate of L/D . Wetted-area ratio is clearly dependent on the actual configuration layout. Figure 3.5 shows a spectrum of design approaches and the resulting wetted-area ratios.

As stated above, L/D depends primarily on the wing span and the wetted area. This suggests a new parameter, the "Wetted Aspect Ratio," which is defined as the wing span squared divided by the total aircraft wetted area. This is very similar to the aspect ratio except that it considers total wetted area instead of wing reference area.

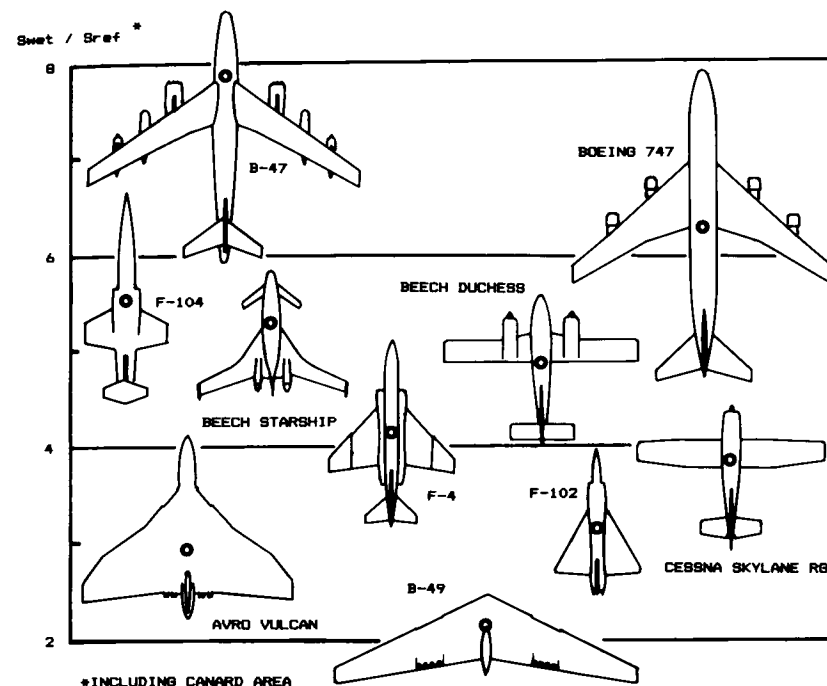


Fig. 3.5 Wetted area ratios.

Figure 3.6 plots maximum L/D for a number of aircraft vs the wetted aspect ratio, and shows clear trend lines for jet, prop, and fixed-gear prop aircraft. Note that the wetted aspect ratio can be shown to equal the wing geometric aspect ratio divided by the wetted-area ratio, S_{wet}/S_{ref} .

It should be clear at this point that the designer has control over the L/D . The designer picks the aspect ratio and determines the configuration arrangement, which in turn determines the wetted-area ratio.

However, the designer must strike a compromise between the desire for a high L/D and the conflicting desire for low weight. The statistical equations provided above for estimating the empty-weight fraction are based on "normal" designs. If the aspect ratio selected is much higher than that of other aircraft in its class, the empty-weight fraction would be higher than estimated by these simple statistical equations.

L/D can now be estimated from a conceptual sketch. This is the crude, "back of a napkin" drawing mentioned earlier. On the conceptual sketch the designer arranges the major components of the aircraft, including wings, tails, fuselage, engines, payload or passenger compartment, landing gear, fuel tanks, and others as needed.

From the sketch the wetted-area ratio can be "eyeball-estimated" using Fig. 3.5 for guidance. The wetted aspect ratio can then be calculated as the wing aspect ratio divided by the wetted-area ratio. Figure 3.6 can then be used to estimate the maximum L/D .

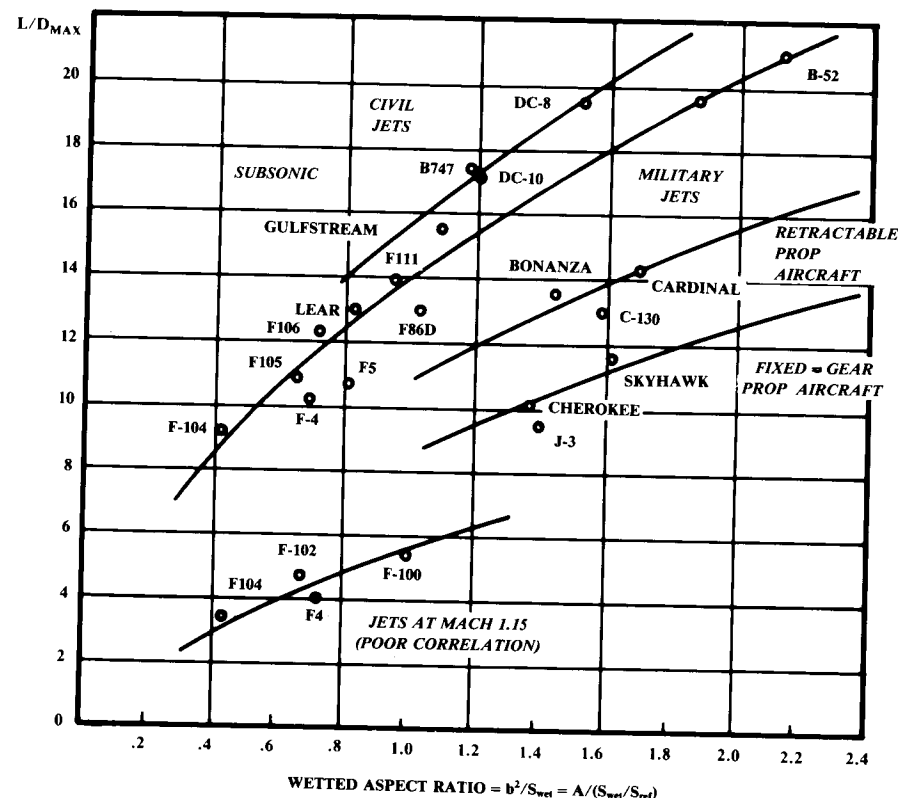


Fig. 3.6 Maximum lift to drag ratio trends.

Note that the L/D can usually be estimated without a sketch by an experienced designer. The wetted aspect ratio method is provided primarily for the student, but can be useful for quickly evaluating novel concepts.

Drag varies with altitude and velocity. For any altitude there is a velocity which maximizes L/D . To maximize cruise or loiter efficiency the aircraft should fly at approximately the velocity for maximum L/D .

For reasons which will be derived later, the most efficient loiter for a jet aircraft occurs exactly at the velocity for maximum L/D , but the most efficient loiter speed for a propeller aircraft occurs at a slower velocity that yields an L/D of 86.6% of the maximum L/D .

Similarly, the most efficient cruise velocity for a propeller aircraft occurs at the velocity yielding maximum L/D , whereas the most efficient cruise for a jet aircraft occurs at a slightly higher velocity yielding an L/D of 86.6% of the maximum L/D .

	Cruise	Loiter
Jet	$0.866 L/D_{max}$	L/D_{max}
Prop	L/D_{max}	$0.866 L/D_{max}$

For initial sizing, these percents can be multiplied times the maximum L/D as estimated using Fig. 3.6 to determine the L/D for cruise and loiter.

Fuel-Fraction Estimation

Using historical values from Table 3.2 and the equations for cruise and loiter segments, the mission-segment weight fractions can now be estimated. By multiplying them together, the total mission weight fraction, W_x/W_0 , can be calculated.

Since this simplified sizing method does not allow mission segments involving payload drops, all weight lost during the mission must be due to fuel usage. The mission fuel fraction must therefore be equal to $(1 - W_x/W_0)$. If you assume, typically, a 6% allowance for reserve and trapped fuel, the total fuel fraction can be estimated as in Eq. (3.11).

$$\frac{W_f}{W_0} = 1.06 \left(1 - \frac{W_x}{W_0} \right) \quad (3.11)$$

3.5 TAKEOFF-WEIGHT CALCULATION

Using the fuel fraction found with Eq. (3.11) and the statistical empty-weight equation selected from Table 3.1, the takeoff gross weight can be found iteratively from Eq. (3.4). This is done by guessing the takeoff gross weight, calculating the statistical empty-weight fraction, and then calculating the takeoff gross weight. If the result doesn't match the guess value, a value between the two is used as the next guess. This will usually converge in just a few iterations. This first-order sizing process is diagrammed in Fig. 3.7.

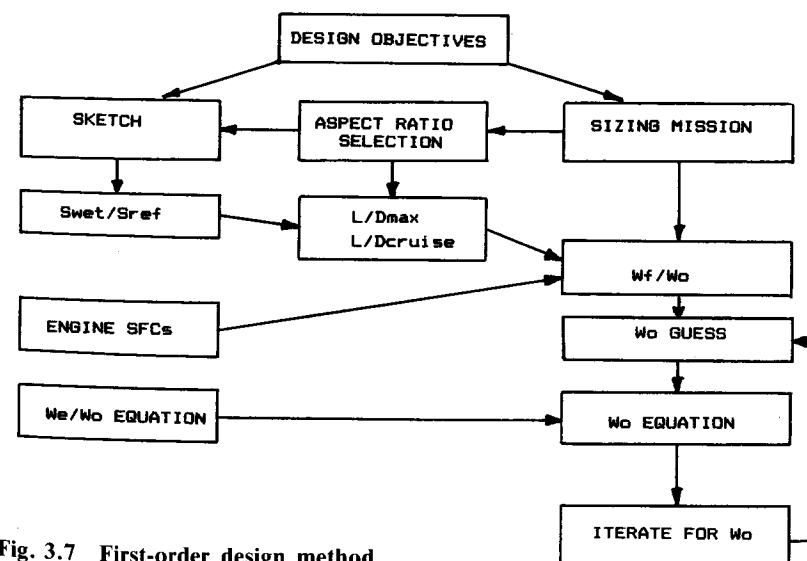


Fig. 3.7 First-order design method.

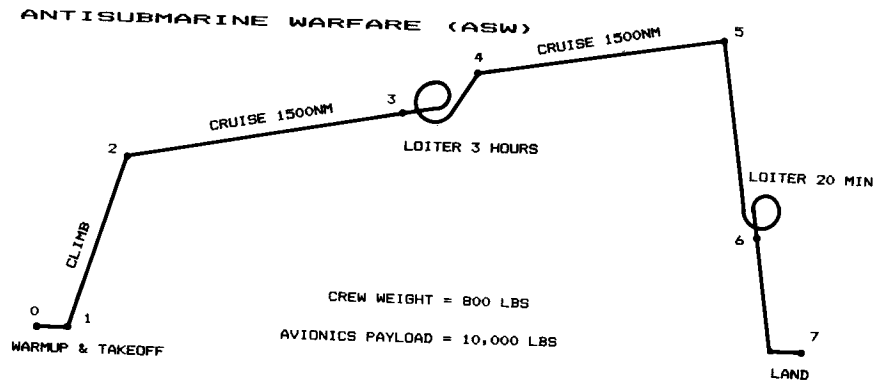


Fig. 3.8 Sample mission profile.

3.6 DESIGN EXAMPLE: ASW AIRCRAFT

ASW Requirements

Figure 3.8 illustrates the mission requirement for a hypothetical antisubmarine warfare (ASW) aircraft. The key requirement is the ability to loiter for three hours at a distance of 1500 n.mi. from the takeoff point. While loitering on-station, this type of aircraft uses sophisticated electronic equipment to detect and track submarines. For the sizing example, this equipment is assumed to weigh 10,000 lb. Also, a four-man crew is required, totalling 800 lb. The aircraft must cruise at 0.6 Mach number.

Conceptual Sketches

Figure 3.9 shows four conceptual approaches considered by the designer in response to these mission requirements. Concept one is the conventional

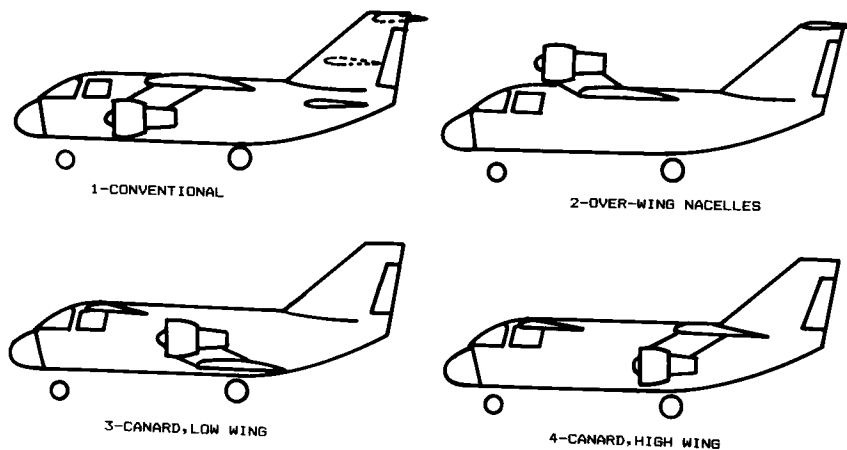


Fig. 3.9 ASW concept sketches.

approach, looking much like the Lockheed S-3A that currently performs a similar mission. The low horizontal tail position shown in solid line would offer the lightest structure, but may place the tail in the exhaust stream of the engines, so other positions for the horizontal tail are shown in dotted lines.

The second concept is much like the first except for the engine location. Here the engines are shown mounted over the wing. This provides extra lift due to the exhaust over the wings, and also provides greater ground clearance for the engines, which reduces the tendency of the jet engines to suck up debris. However, the disadvantage of this concept is the difficulty in reaching the engines for maintenance work.

Concepts three and four explore the canarded approach. Canards offer the potential for reduced trim drag and may provide a wider allowable range for the center of gravity. In concept three, the wing is low and the engines are mounted over the wing as in concept two. This would allow the main landing gear to be stowed in the wing root.

In concept four, the wing is high with the engines mounted below. This last approach offers better access to the engines, and for this reason was selected for further development.

Figure 3.10 is a conceptual sketch prepared, in more detail, for the selected concept. Note the locations indicated for the landing-gear stowage, crew station, and fuel tanks.

This points out a common problem with canard aircraft, the fuel tank locations. The fuel tanks should be placed so that the fuel is evenly dis-

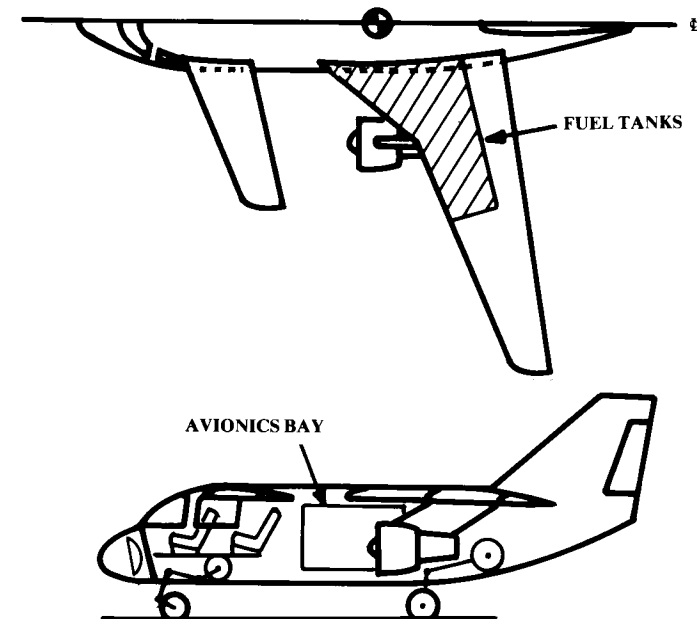


Fig. 3.10 Completed ASW sketch.

tributed about the aircraft center of gravity (estimated location shown by the circle with two quarters shaded). This is necessary so that the aircraft when loaded has nearly the same center of gravity as when its fuel is almost gone. However, the wing is located aft of the center of gravity whenever a canard is used, so the fuel located in the wing is also aft of the center of gravity.

One solution to this problem would be to add fuel tanks in the fuselage, forward of the center of gravity. This would increase the risk of fire in the fuselage during an accident, and is forbidden in commercial aircraft. Although this example is a military aircraft, fire safety should always be considered.

Another solution, shown on the sketch, is to add a wing strake full of fuel. This solution is seen on the Beech Starship among others. The strakes do add to the aircraft wetted area, which reduces cruise aerodynamic efficiency.

This example serves to illustrate an extremely important principle of aircraft design; namely, that there is no such thing as a free lunch! All aircraft design entails a series of trade offs. The canard offers lower trim drag, but in this case may require a higher wetted area. The only true way to determine whether a canard is a good idea for this or any aircraft is to design several aircraft, one with and one without a canard. This type of trade study comprises the majority of the design effort during the conceptual design process.

L/D Estimation

For initial sizing, a wing aspect ratio of about 11 was selected. With the area of the wing and canard both included, this is equivalent to a combined aspect ratio of about 8.

Comparing the sketch of Fig. 3.10 to the examples of Fig. 3.5, it would appear that the wetted area ratio (S_{wet}/S_{ref}) is about 5.5. This yields a wetted aspect ratio of 1.45 (i.e., 8/5.5).

For a wetted aspect ratio of 1.45, Fig. 3.6 indicates that a maximum lift-to-drag ratio of about 17 would be expected. This value, obtained from an initial sketch and the selected aspect ratio, can now be used for initial sizing.

Since this is a jet aircraft, the maximum L/D is used for loiter calculations. For cruise, a value of 0.866 times the maximum L/D , or about 15, is used.

Takeoff-Weight Sizing

From Table 3.3, initial values for SFC are obtained. For a subsonic aircraft the best SFC values are obtained with high-bypass turbofans, which have typical values of about 0.5 for cruise and 0.4 for loiter.

Table 3.1 does not provide an equation for statistically estimating the empty weight fraction of an antisubmarine aircraft. However, such an aircraft is basically designed for subsonic cruise efficiency so the equation for military cargo/bomber can be used. The extensive ASW avionics would not be included in that equation, so it is treated as a separate payload weight.

Box 3.1 ASW sizing calculations

Mission Segment Weight Fractions

1) Warmup and takeoff	$W_1/W_0 = 0.97$	(Table 2)
2) Climb	$W_2/W_1 = 0.985$	(Table 2)
3) Cruise	$R = 1500\text{nm} = 9,114,000 \text{ ft}$ $C = 0.5 \text{ 1/hr} = 0.0001389 \text{ 1/s}$ $V = 0.6\text{M} \times (994.8 \text{ ft/s}) = 569.9 \text{ ft/s}$ $L/D = 16 \times 0.866 = 13.9$ $W_3/W_2 = e^{-(RC/VL/D)} = e^{-0.16} = 0.852$	
4) Loiter	$E = 3 \text{ hours} = 10,800 \text{ s}$ $C = 0.4 \text{ 1/hr} = 0.0001111 \text{ 1/s}$ $L/D = 16$ $W_4/W_3 = e^{-(EC/L/D)} = e^{-0.075} = 0.9277$	
5) Cruise (same as 3)	$W_5/W_4 = 0.852$	
6) Loiter	$E = 1/3 \text{ hours} = 1200 \text{ s}$ $C = 0.0001111 \text{ 1/s}$ $L/D = 16$ $W_6/W_5 = e^{-0.0083} = 0.9917$	
7) Land	$W_7/W_6 = 0.995$	(Table 2)

$$W_7/W_0 = (0.97)(0.985)(0.852)(0.9277)(0.852)(0.9917)(0.995) = 0.635$$

$$W_f/W_0 = 1.06(1 - 0.6505) = 0.387$$

$$W_e/W_0 = 0.93 W_0^{-0.07} \quad (\text{Table 1})$$

$$W_0 = \frac{10,800}{1 - 0.387 - \frac{W_e}{W_0}} = \frac{10,800}{0.613 - 0.93 W_0^{-0.07}}$$

W_0 Guess	W_e/W_0	W_0 Calculated
50,000	0.4361	61,057
60,000	0.4305	59,191
59,200	0.4309	59,328
59,300	0.4309	59,311
59,310	0.4309	59,309.6

Box 3.1 gives the calculations for sizing this example. Note the effort to insure consistent dimensions, including the conversion of cruise velocity (Mach 0.6) to ft/s by assuming a typical cruise altitude of 30,000 ft. At this altitude the speed of sound (see Appendix A.2) is 994.8 ft/s.

The calculations in box 3.1 indicate a takeoff gross weight of 59,310 lb. Although these calculations are based upon crude estimates of aerodynamics, weights, and propulsion parameters, it is interesting to note that the actual takeoff gross weight of the Lockheed S-3A, as quoted in Ref. 1, is 52,539. While strict accuracy should not be expected, this simple sizing method will usually yield an answer in the "right ballpark."

Trade Studies

An important part of conceptual design is the evaluation and refinement, with the customer, of the design requirements. In the ASW design example, the required range of 1500 n.mi. (each way) is probably less than the customer would really like. A "range trade" can be calculated to determine the increase in design takeoff gross weight if the required range is increased.

Box 3.2 Range trade

1000 nm Range

$$W_3/W_2 = W_5/W_4 = e^{-0.1065} = 0.899$$

$$W_7/W_0 = 0.7069$$

$$W_f/W_0 = 1.06(1 - 0.7069) = 0.3107$$

$$W_0 = \frac{10,800}{1 - 0.3107 - \frac{W_e}{W_0}} = \frac{10,800}{0.6893 - 0.93W_0^{-0.07}}$$

W_0 Guess	W_e/W_0	W_0 Calculated
50,000	0.4361	42,657
45,000	0.4393	43,203
43,500	0.4403	43,384
43,400	0.4404	43,396
43,398	0.4404	43,397

2000 nm Range

$$W_3/W_2 = W_5/W_4 = e^{-0.213} = 0.8082$$

$$W_7/W_0 = 0.5713$$

$$W_f/W_0 = 0.4544$$

$$W_0 = \frac{10,800}{1 - 0.4544 - \frac{W_e}{W_0}} = \frac{10,800}{0.5456 - 0.93W_0^{-0.07}}$$

W_0 Guess	W_e/W_0	W_0 Calculated
50,000	0.4361	98,660
80,000	0.4219	87,331
86,000	0.4198	85,889
85,900	0.4199	85,913
85,910	0.4199	85,911

This is done by recalculating the weight fractions for the cruise mission segments, using arbitrarily selected ranges. For example, instead of the required 1500 n.mi., we will calculate the cruise weight fractions using 1000 and 2000 n.mi., and will size the aircraft separately for each of those ranges. These calculations are shown in Box 3.2, and the results are plotted in Fig. 3.11.

In a similar fashion, a "payload trade" can be made. The mission-segment weight fractions and fuel fraction are unchanged but the numerator of the sizing equation, Eq. (3.4), is parametrically varied by assuming different payload weights. The given payload requirement is 10,000 lb of avionics equipment. Box 3.3 shows the sizing calculations assuming payload weights of 5000, 15,000, and 20,000 lb. The results are plotted in Fig. 3.12.

The statistical empty-weight equation used here for sizing was based upon existing military cargo and bomber aircraft, which are all of aluminum construction. The takeoff gross weight calculations above have thus implicitly assumed that the new aircraft would also be built of aluminum.

To determine the effect of building the aircraft out of composite materials, the designer must adjust the empty-weight equation. As previously mentioned, this can be approximated in the early stages of design by taking 95% of the empty-weight fraction obtained for a metal aircraft. The calculations for resizing the aircraft using composite materials are shown in box 3.4.

The use of composite materials reduces the takeoff gross weight from 59,310 lb to only 53,771 lb yet the aircraft can still perform the same

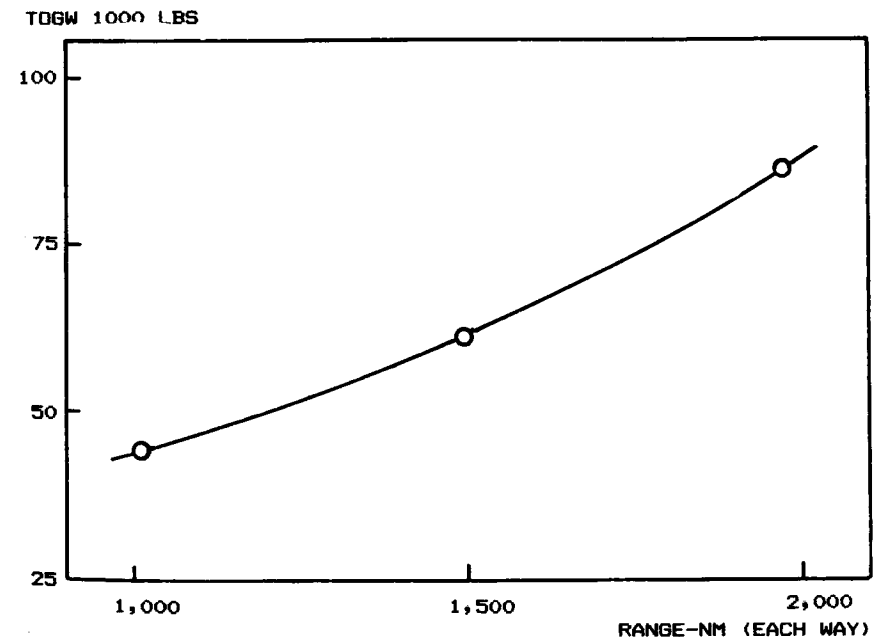


Fig. 3.11 Range trade.

Box 3.3 Payload trade

$$\text{Payload} = 5000 \text{ lb} \quad W_0 = \frac{5800}{0.613 - 0.93 W_0^{-0.07}}$$

W_0 Guess	W_e / W_0	W_0 Calculated
50,000	0.4361	32,787
35,000	0.4471	34,960
34,970	0.4471	34,965
34,966	0.4471	34,966

$$\text{Payload} = 15,000 \text{ lb} \quad W_0 = \frac{15,800}{0.613 - 0.93 W_0^{-0.07}}$$

W_0 Guess	W_e / W_0	W_0 Calculated
50,000	0.4362	89,366
85,000	0.4202	81,937
82,000	0.4212	82,389
82,350	0.4211	82,335

$$\text{Payload} = 20,000 \text{ lb} \quad W_0 = \frac{20,800}{0.613 - 0.93 W_0^{-0.07}}$$

W_0 Guess	W_e / W_0	W_0 Calculated
90,000	0.4185	106,941
100,000	0.4154	105,272
105,000	0.4140	104,522
104,600	0.4141	104,581

Box 3.4 Composite material trade

$$W_e / W_0 = (0.95)(0.93 W_0^{-0.07}) = 0.8835 W_0^{-0.07}$$

$$W_0 = \frac{10,800}{1 - 0.387 - \frac{W_e}{W_0}} = \frac{10,800}{0.613 - 0.8835 W_0^{-0.07}}$$

W_0 Guess	W_e / W_0	W_0 Calculated
50,000	0.4143	54,344
54,000	0.4120	53,742
53,800	0.4122	53,771

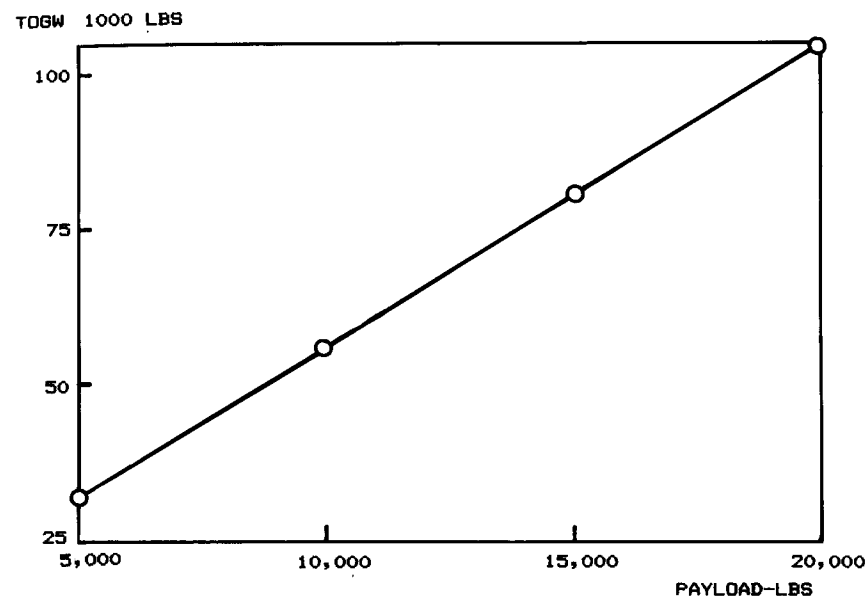


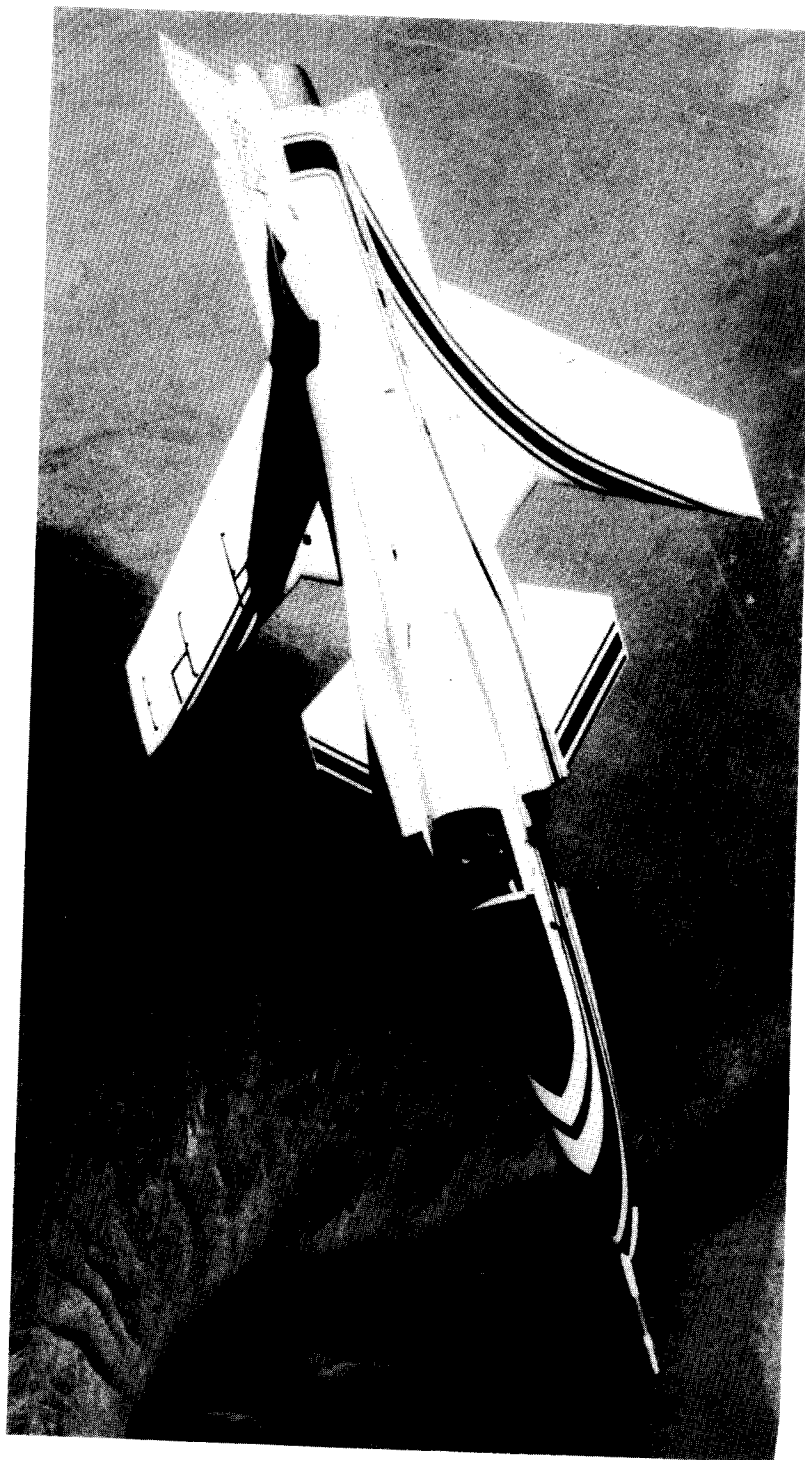
Fig. 3.12 Payload trade.

mission. This is a 9.3% takeoff-weight savings, resulting from only a 5% empty-weight saving.

This result sounds erroneous, but is actually typical of the “leverage” effect of the sizing equation. Unfortunately, this works both ways. If the empty weight creeps up during the detail-design process, it will require a more-than-proportional increase in takeoff gross weight to maintain the capability to perform the sizing mission. Thus it is crucial that realistic estimates of empty weight be used during early conceptual design, and that the weight be strictly controlled during later stages of design.

There are many trade studies which could be conducted other than range, payload, and material. Methods for trade studies are discussed in detail in Chapter 19.

The remainder of the book presents better methods for design, analysis, sizing, and trade studies, building on the concepts just given. In this chapter a conceptual sketch was made, but no guidance was provided as to how to make the sketch or why different features may be good or bad. Following chapters address these issues and illustrate how to develop a complete three-view drawing for analysis. Then more-sophisticated methods of analysis, sizing, and trade studies will be provided.



X-29 Forward-Swept-Wing Demonstrator

AIRFOIL AND GEOMETRY SELECTION

4.1 INTRODUCTION

Before the design layout can be started, values for a number of parameters must be chosen. These include the airfoil(s), the wing and tail geometries, wing loading, thrust-to-weight or horsepower-to-weight ratio, estimated takeoff gross weight and fuel weight, estimated wing, tail, and engine sizes, and the required fuselage size. These are discussed in the next three chapters.

This chapter covers selecting the airfoil and the wing and tail geometry. Chapter 5 addresses estimation of the required wing loading and thrust-to-weight ratio (horsepower-to-weight ratio for a propeller aircraft). Chapter 6 provides a more refined method for initial sizing than the quick method presented in the last chapter, and concludes with the use of the sizing results to calculate the required wing and tail area, engine size, and fuselage volume.

4.2 AIRFOIL SELECTION

The airfoil, in many respects, is the heart of the airplane. The airfoil affects the cruise speed, takeoff and landing distances, stall speed, handling qualities (especially near the stall), and overall aerodynamic efficiency during all phases of flight.

Much of the Wright Brothers' success can be traced to their development of airfoils using a wind tunnel of their own design, and the in-flight validation of those airfoils in their glider experiments of 1901-1902. The P-51 was regarded as the finest fighter of World War II in part because of its radical laminar-flow airfoil. Recently, the low-speed airfoils developed by Peter Lissaman contributed to the success of the man-powered Gossamer Condor.

Airfoil Geometry

Figure 4.1 illustrates the key geometric parameters of an airfoil. The front of the airfoil is defined by a leading-edge radius which is tangent to the upper and lower surfaces. An airfoil designed to operate in supersonic flow will have a sharp or nearly-sharp leading edge to prevent a drag-producing bow shock. (As discussed later, wing sweep may be used instead of a sharp leading edge to reduce the supersonic drag.)

The chord of the airfoil is the straight line from the leading edge to the trailing edge. It is very difficult to build a perfectly sharp trailing edge, so most airfoils have a blunt trailing edge with some small finite thickness.